

Roll No: CS-11 / BT230011CS. BT/CS/S-26/CS6E003C/03
Aman Sheikh. VSODG - DIP
(D7) (Regular)



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(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU & RTMNU



B. Tech Summer / Winter - 2026
(ESE / Make-up / Supplementary) Examination

Branch: Computer Science & Engineering
Subject: PEC-III (Digital Image Processing)
Marks: 100

Sem: - VI
Subject Code: CS6E003C
Date of Exam: 30/04/26 Time: 3 Hrs.

Instructions to the students:

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat sketches.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

COs: Students should be able to:

- CO 1: Understand the fundamental concepts of a digital image processing system.
CO 2: Analyze images in the frequency domain using various transforms
CO 3: Evaluate the techniques for image enhancement and image restoration.
CO 4: Categorize various compression techniques.
CO 5: Interpret Image compression standards.

SECTION A

1. All Questions are compulsory.
2. Answer in one/two word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Digital Image.	L1/CO1	2
Q.2	List any two features of MATLAB.	L1/CO2	2
Q.3	Discuss in short Image As Data.	L3/CO1	2
Q.4	Define matrix in MATLAB.	L2/CO2	2
Q.5	Write short note on pixel.	L3/CO1	2
Q.6	Define image enhancement.	L1/CO3	2
Q.7	Identify the Image Processing Toolbox.	L1/CO2	2
Q.8	State the Measures of Image.	L1/CO1	2
Q.9	Identify the Different Morphological Algorithm.	L2/CO5	2
Q.10	State the levels of Image Processing.	L1/CO1	2
Q.11	Write the Components of Digital Image.	L3/CO1	2
Q.12	Define Image Morphology.	L1/CO1	2

SECTION B

1. All Questions are compulsory.

Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Illustrate different types of image representation.	L4/CO1	4
Q.14	Explain the JPEG compression standard.	L3/CO4	4
Q.15	Describe different data types supported by MATLAB.	L3/CO2	4
Q.16	Discuss basic Gray level transformation functions.	L2/CO3	4
Q.17	Examine the need for data compression in images.	L3/CO4	4
Q.18	Analyse different colour models used in image processing.	L3/CO5	4

SECTION C

1. Solve any FOUR Questions.

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Discuss about Converting colors from HSI to RGB	L4/CO5	7
Q.20	Explain image enhancement using arithmetic and logical operations.	L5/CO3	7
Q.21	Illustrate the fundamentals of digital image processing.	L5/CO1	7
Q.22	Describe the Run Length Encoding for compressing string and solve this example: a) BBBB BBBB AAAA ANNN NNNG GG GG MMM b) 0000011111000000111111000000111111000000111111	L4/CO4	7
Q.23	Analyse File I/O operations in MATLAB with suitable commands.	L5/CO2	7
Q.24	Examine the Huffman coding for the probability of 0.30, 0.25, 0.20, 0.12, 0.08, 0.05 and find out the average codeword length, Entropy & Efficiency.	L5/CO4	7

SECTION D

1. Solve any TWO Questions.

Q. No.	Questions	Level/CO Mapped	Marks																		
Q.25	Describe the applications of digital image processing in different fields such as medicine, satellite imaging, and security.	L5/CO1	12																		
Q.26	Examine the Histogram Equalizer for given below are two histograms (a) and (b), modify the histogram (a) as given by histogram (b).	L5/CO3	12																		
(a)	<table><tr><td>Gray Levels</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>No. of Pixels</td><td>80</td><td>100</td><td>90</td><td>60</td><td>30</td><td>20</td><td>10</td><td>0</td></tr></table>	Gray Levels	0	1	2	3	4	5	6	7	No. of Pixels	80	100	90	60	30	20	10	0		
Gray Levels	0	1	2	3	4	5	6	7													
No. of Pixels	80	100	90	60	30	20	10	0													
(b)	<table><tr><td>Gray Levels</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>No. of Pixels</td><td>0</td><td>0</td><td>0</td><td>60</td><td>80</td><td>100</td><td>80</td><td>70</td></tr></table>	Gray Levels	0	1	2	3	4	5	6	7	No. of Pixels	0	0	0	60	80	100	80	70		
Gray Levels	0	1	2	3	4	5	6	7													
No. of Pixels	0	0	0	60	80	100	80	70													
Q.27	Explain smoothing, sharpening, and segmentation of colour images in detail.	L5/CO5	12																		

8 4 2 1

Q3-28%

(75)